

INTERNATIONAL ISLAMIC UNIVERSITY CHITTAGONG

Department of Computer Science and Engineering

CSE-2340: Software Development 1

Semantic HTML

Week 1 · Class 3 · Headings, text, links, images, forms, document structure

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Recall & Today's Outcome

? Recall from Class 2

- What does the browser do with an HTML file?
- Which extension did we use for our first page?
- How do we open a page with Live Server?
- Where does the page title text appear?

✓ By the end of today you can

- Explain why semantic HTML matters
- Build a valid HTML5 document structure
- Use headings, text, links, images and lists
- Use header, nav, main, section, footer
- Build a simple form with labels and inputs
- Publish a complete personal profile page

CONCEPT

What is Semantic HTML?

Semantic means "with meaning". A semantic tag tells the browser what the content IS, not how it looks.

NOT SEMANTIC

```
<div class="top">
  <div class="big-text">My Profile</div>
</div>
<div class="middle">
  <div>Hello, I am a student.</div>
</div>
```

SEMANTIC

```
<header>
  <h1>My Profile</h1>
</header>
<main>
  <p>Hello, I am a student.</p>
</main>
```

Both look the same in the browser. Only one is understandable.

- Screen readers can jump straight to the main content for a blind user
- Google understands which text is the real heading of your page
- Your teammate reads the code in five seconds instead of five minutes
- Analogy: a building with rooms labelled Kitchen and Bedroom, not Room1 and Room2

Anatomy of an HTML Document

```
<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="UTF-8">
    <title>My Profile</title>
  </head>
  <body>
    <h1>Hello</h1>
  </body>
</html>
```

<!DOCTYPE html>

Tells the browser: this is HTML5

<html lang="en">

The whole page. lang helps screen readers

<head>

Information ABOUT the page. Not visible

<title>

Text shown on the browser tab

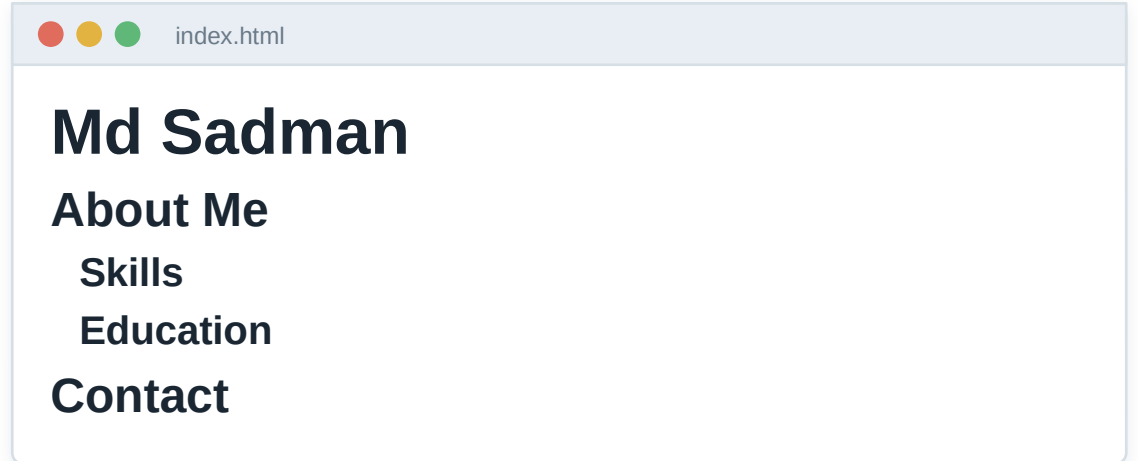
<body>

Everything the user actually sees

Headings: `<h1>` to `<h6>`

One h1 per page. Never skip a level. Headings form the table of contents of your page.

```
<h1>Md Sadman</h1>
<h2>About Me</h2>
<h3>Skills</h3>
<h3>Education</h3>
<h2>Contact</h2>
```



Common mistake

- Using h1 five times because it looks bigger
- Jumping from h1 straight to h4
- Using a heading only to make text bold

Correct habit

- h1 = page title, used once
- h2 = major sections
- h3 = subsections inside h2

Text Elements

```
<p>I am a CSE student at IIUC.</p>
```

```
<p>I am <strong>learning</strong> web  
development and I  
<em>enjoy</em> it.</p>
```

```
<p>Line one<br>Line two</p>
```

```
<hr>
```

Browser output

I am a CSE student at IIUC.

I am **learning** web development and I *enjoy* it.

Line one
Line two

<p>

A paragraph of text

Important - shown bold

Emphasis - shown italic

**
**

Line break, no closing tag

<hr>

Thematic break line

Links: the `<a>` Element

```
<!-- External link: goes to another website -->
<a href="https://www.iiuc.ac.bd">Visit IIUC</a>

<!-- Internal link: goes to another file in my project -->
<a href="about.html">About Me</a>

<!-- Anchor link: jumps to a section on the same page -->
<a href="#contact">Contact Me</a>
```

1 href

- The address the link opens
- Missing href = not a real link

2 Link text

- Write what the user will get
- Never write "click here"

3 id target

- #contact needs id="contact"
- on the section you jump to

Images: the Element

```

```

```
<!-- src  = where the file is
      alt  = text if the image fails or
             for a screen reader user
      img has NO closing tag -->
```



Why alt text matters

- A blind user hears the alt text
- It shows if the file path is wrong
- Search engines read it
- Decorative image? Use alt=""

PATH RULES YOU WILL NEED TODAY

profile.jpg

Same folder as index.html

images/profile.jpg

Inside the images folder

../profile.jpg

One folder above

https://.../pic.jpg

A file on the internet

Lists: `<ul>`, `<ol>`, `<li>`

```
<!-- Order does not matter -->
<ul>
  <li>HTML</li>
  <li>CSS</li>
  <li>Java</li>
</ul>

<!-- Order matters -->
<ol>
  <li>Open VS Code</li>
  <li>Create index.html</li>
  <li>Run Live Server</li>
</ol>
```

Browser output

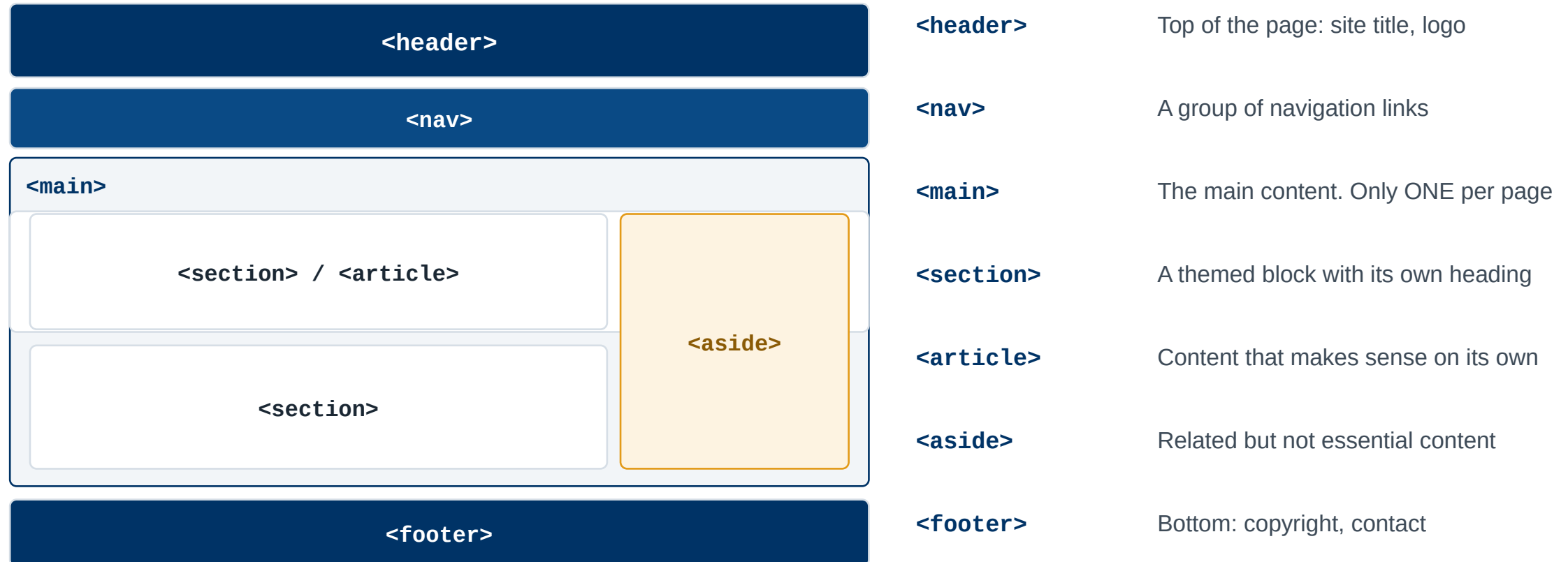
- HTML
 - CSS
 - Java
-
1. Open VS Code
 2. Create index.html
 3. Run Live Server

Rule: only `<li>` can be a direct child of `<ul>` or `<ol>`. Put paragraphs and links **INSIDE the `<li>`.**

CONCEPT

Document Structure Elements

These tags replace endless `<div>` tags. Each one names a region of the page.



Forms: Collecting User Input

```
<form>
  <label for="name">Your Name</label>
  <input type="text" id="name" name="name">

  <label for="email">Your Email</label>
  <input type="email" id="email" name="email">

  <label for="msg">Message</label>
  <textarea id="msg" name="msg"></textarea>

  <button type="submit">Send</button>
</form>
```



The golden rule

- Every input needs a <label>
- label for="x" must match input id="x"
- Clicking the label focuses the input

Useful input types

- text - normal typing
- email - checks for @
- password - hides characters
- number, date, checkbox, radio

01

Live Build: My Personal Profile Page

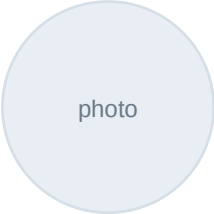
We build it together in four steps. Open your empty `index.html` now.

What We Are Building

index.html - Live Server

Md Sadman Hafiz

[Home](#) [About](#) [Skills](#) [Contact](#)



About Me

I am a third semester CSE student at IIUC.
I am learning front-end development.

Skills

- HTML
- CSS basics
- Java

Contact Me

Name

Email

© 2026 Md Sadman Hafiz

Regions we will write

- header + nav - name and menu
- main - profile, skills, education
- form - name, email, message
- footer - copyright

→ Before we start

- Folder: my-profile
- File: index.html
- Folder: images (put any photo)
- Run with Live Server

Step 1: The Skeleton

```
<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="UTF-8">
    <meta name="viewport"
      content="width=device-width, initial-scale=1.0">
    <title>Md Sadman | Profile</title>
  </head>
  <body>

    <!-- everything goes here -->

  </body>
</html>
```

1 Fast way

- Type ! then press Tab in VS Code
- VS Code writes the skeleton for you
- Change only the <title>

✓ Check now

- Save with Ctrl + S
- Right click → Open with Live Server
- Your tab shows the title

Step 2: Header and Navigation

```
<header>
  <h1>Md Sadman Hafiz</h1>
  <p>Third semester CSE student, IIUC</p>

  <nav>
    <ul>
      <li><a href="#about">About</a></li>
      <li><a href="#skills">Skills</a></li>
      <li><a href="#contact">Contact</a></li>
    </ul>
  </nav>
</header>
```

Read it out loud

- header holds the page identity
- nav holds a ul of links
- Each link is one li

? Why # links

- #about jumps to id="about"
- We add those ids in Step 3
- Until then the link does nothing

Step 3: Main Content

```
<main>
  <section id="about">
    <h2>About Me</h2>
    
    <p>I am a <strong>third semester</strong> CSE student at IIUC. I am learning
      front-end development and I <em>enjoy</em> building web pages.</p>
  </section>

  <section id="skills">
    <h2>Skills</h2>
    <ul>
      <li>HTML</li>
      <li>Java</li>
      <li>Problem solving</li>
    </ul>
  </section>

  <section id="education">
    <h2>Education</h2>
    <ol>
      <li>BSc in CSE, IIUC - running</li>
      <li>Higher Secondary Certificate - 2024</li>
    </ol>
  </section>
</main>
```


Step 4: Form and Footer

```
<section id="contact">
  <h2>Contact Me</h2>
  <form>
    <label for="name">Name</label>
    <input type="text" id="name" name="name" placeholder="Your full name">

    <label for="email">Email</label>
    <input type="email" id="email" name="email" placeholder="you@example.com">

    <label for="message">Message</label>
    <textarea id="message" name="message" rows="4"></textarea>

    <button type="submit">Send Message</button>
  </form>
</section>
</main>

<footer>
  <p>&copy; 2026 Md Sadman Hafiz. Built for CSE-2340.</p>
</footer>
```

Accessibility Checklist



Landmarks

header, nav, main, footer are present. Exactly one main.



Labels

Every input has a label and the for matches the id.



Link text

Link text describes the destination, not "click here".



Alt text

Every img has a useful alt describing the picture.



Heading order

One h1, then h2, then h3. No skipped level.



Language

html lang="en" is set so screen readers pronounce correctly.

Common Mistakes and Fixes

Image does not appear

Wrong path or wrong file name. Check images/profile.jpg letter by letter. File names are case sensitive.

Nav link does nothing

The id on the section is missing or spelled differently from the href.

Page looks like one long line

A tag was never closed. Look for a missing `</div>` or `</section>`.

Nothing changes in the browser

The file was not saved, or Live Server is showing a different file.

Debug method: read the error out loud, check the smallest thing first (spelling), then the next smallest (path).

Assignment - Class 3

1 Part 1 - In-class demonstration

- Time: last 5 minutes of this class
- Task: add one new `<section id="hobbies">` with an `h2` and a `<ul>` of three hobbies
- Add a matching link in the nav
- Show the working jump link on your screen
- Submission: show to the instructor before leaving
- Marks: 2 (checked on the spot)

2 Part 2 - Home task

- Complete the full profile page from today
- Add an `<aside>` with one favourite quote
- Add a second page `about.html` and link it from the nav
- File name: `index.html` and `about.html` in one folder
- Deadline: before the next class (Week 2, Class 1)
- Submission: bring the folder on your laptop; we push it to GitHub next class
- Marks: 3

Keep this folder safe. In Class 4 it becomes your first Git repository and your first GitHub push.

48-50 MINUTES

Marking Rubric & Exit Check

RUBRIC - 5 MARKS TOTAL

Valid HTML5 structure and correct nesting	1.5
Semantic regions used (header, nav, main, footer)	1.5
Headings, lists, image with alt, working links	1.0
Form with labels correctly linked to inputs	1.0

Exit question 1 - concept

Why is `<header>` better than `<div class="header">`?

Exit question 2 - practical

Write the one line of HTML that shows an image named `logo.png` stored in the `images` folder, with correct alt text.

Next class: Git basics - we turn today's folder into a repository and push it to GitHub.